

Data Mining

Lab - 5

¥@\$# Kakadiya | 23010101121 | 2/7/2025

Lab - 5 - Data Preprocessing

1) First, you need to read the titanic dataset from local disk and display Last five records

```
In [17]: import pandas as pd
```

```
In [18]: df=pd.read_csv('../data/titanic.csv')
df
```

Out[18]:

	PassengerId	Survived	Pclass		Name	Sex	Age	SibSp	Parch		Ticket	Fare	Cabin	Embarked
	0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0		A/5 21171	7.2500	NaN	S
	1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0		PC 17599	71.2833	C85	C
	2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0		STON/O2. 3101282	7.9250	NaN	S
	3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0		113803	53.1000	C123	S
	4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0		373450	8.0500	NaN	S
	...	...	...	...	...	...	...	...	...		...	...	...	...
	886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0		211536	13.0000	NaN	S
	887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0		112053	30.0000	B42	S
	888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2		W./C. 6607	23.4500	NaN	S
	889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0		111369	30.0000	C148	C
	890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0		370376	7.7500	NaN	Q

891 rows × 12 columns

```
In [19]: df.tail()
```

Out[19]:

	PassengerId	Survived	Pclass		Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2		Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1		Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3		Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1		Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3		Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

2) Handle Missing Values in data set [use dropna(), fillna(), and interpolate]

In [20]:

```
# using dropna

df_dropna = df.dropna()
# df_dropna = df.dropna(how = 'all')
# df_dropna = df.dropna(how = 'any', axis = 1)
df_dropna
```

Out[20]:

	PassengerId	Survived	Pclass		Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	2	1	1		Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
3	4	1	1		Futelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
6	7	0	1		McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
10	11	1	3		Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000	G6	S
11	12	1	1		Bonnell, Miss. Elizabeth	female	58.0	0	0	113783	26.5500	C103	S
...	...	...	...		...	...	...	...	...	...	...	...	...
871	872	1	1		Beckwith, Mrs. Richard Leonard (Sallie Monypeny)	female	47.0	1	1	11751	52.5542	D35	S
872	873	0	1		Carlsson, Mr. Frans Olof	male	33.0	0	0	695	5.0000	B51 B53 B55	S
879	880	1	1		Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)	female	56.0	0	1	11767	83.1583	C50	C
887	888	1	1		Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
889	890	1	1		Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C

183 rows × 12 columns

In [21]:

```
# using fillna
df_fillna=df.copy()
# df_fillna = df.fillna(30)
# df_fillna = df.fillna({'Age': 35, 'Cabin': 'C56'})
age_mean = df.Age.mean()
df_fillna = df.fillna({'Age':age_mean})
```

In [22]:

```
# using interpolate

data_interpolate = df.interpolate()
data_interpolate['Age']= data_interpolate['Age'].interpolate()
data_interpolate
```

Out[22]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
	0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
	1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
	2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
	3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
	4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
	...	...	...	...	...	...	...	...	...	...	...	...	...
	886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
	887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
	888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	22.5	1	2	W./C. 6607	23.4500	NaN	S
	889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
	890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

3) Apply Scaling to AGE attribute with min max, decimal scaling and z score.

In [23]:

```
scaling_df=df.copy()
scaling_df['Age']=scaling_df['Age'].interpolate()
df_min_max=scaling_df.copy()
mmAge=df_min_max['Age']
minAge=mmAge.min()
maxAge=mmAge.max()
df_min_max['MinMaxAge']=(mmAge-minAge)/(maxAge-minAge)
df_min_max
```

Out[23]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	MinMaxAge
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	0.271174
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	0.472229
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	0.321438
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	0.434531
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	0.434531
...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S	0.334004
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S	0.233476
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	22.5	1	2	W./C. 6607	23.4500	NaN	S	0.277457
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C	0.321438
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q	0.396833

891 rows × 13 columns